

**36.0.1 ISRO CSE 2008 | Question: 53**

Thrashing

- A. always occurs on large computers
- B. is a natural consequence of virtual memory systems
- C. can always be avoided by swapping
- D. can be caused by poor paging algorithms

isro2008 operating-system paging

Answer key 

36.0.2 ISRO CSE 2008 | Question: 52

Dynamic address translation

- A. is part of the operating system paging algorithm
- B. is useless when swapping is used
- C. is the hardware necessary to implement paging
- D. storage pages at a specific location on disk

isro2008 operating-system paging

Answer key 

36.0.3 ISRO CSE 2009 | Question: 8

The primary purpose of an operating system is

- A. To make most efficient use of the computer hardware
- B. To allow people to use the computer
- C. To keep systems programmers employed
- D. To make computers easier to use

isro2009 operating-system

Answer key 

36.0.4 ISRO CSE 2007 | Question: 25

What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

- A. Partitioning
- B. Multi-tasking
- C. Windowing
- D. Paging

isro2007 operating-system paging

Answer key 

36.0.5 ISRO CSE 2008 | Question: 54

What is the name of the operating system that reads and reacts in terms of operating system?

- A. Batch system
- B. Quick response time
- C. real time system
- D. Time sharing system

Answer key **36.0.6 ISRO CSE 2008 | Question: 58**

Checkpointing a job

- A. allows it to be completed successfully
- B. allows it to continue executing later
- C. prepares it for finishing
- D. occurs only when there is an error in it

Answer key **36.0.7 ISRO CSE 2009 | Question: 11**

A page fault

- A. Occurs when a program accesses an available page on memory
- B. is an error in a specific page
- C. is a reference to a page belonging to another program
- D. occurs when a program accesses a page not currently in memory

Answer key **36.0.8 ISRO CSE 2017 | Question: 52**

Given reference to the following pages by a program

0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,3,8,3

How many page faults will occur if the program has three page frames available to it and uses an optimal replacement?

- A. 7
- B. 8
- C. 9
- D. None of the above

Answer key **36.0.9 ISRO CSE 2018 | Question: 20**

Determine the number of page faults when references to pages occur in the order - 1,2,4,5,2,1,2,4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 having brought earlier than page 2. (assume LRU i.e. Least-Recently-Used algorithm is applied)

- a. 3
- b. 4
- c. 5
- d. None of the above

Answer key **36.0.10 ISRO CSE 2023 | Question: 58**

In which of the virtualization techniques, hypervisor modifies the guest OS?

- A. Full virtualization
- B. Para virtualization
- C. Both (a) and (b)
- D. None of the above

Answer key 

36.1

Bankers Algorithm (1)

36.1.1 Bankers Algorithm: ISRO CSE 2017 | Question: 48



What problem is solved by Dijkstra banker' algorithm?

- A. Mutual exclusion
- B. Deadlock recovery
- C. Deadlock avoidance
- D. Cache coherence

isro2017 operating-system bankers-algorithm deadlock-prevention-avoidance-detection easy

Answer key 

36.2

Concurrency (1)

36.2.1 Concurrency: ISRO-2013-62



A particular parallel program computation requires 100 seconds when executed on a single CPU. If 20% of this computation is strictly sequential, then theoretically the best possible elapsed times for this program running on 2 CPUs and 4 CPUs respectively are

- A. 55 and 45 seconds
- B. 80 and 20 seconds
- C. 75 and 25 seconds
- D. 60 and 40 seconds

isro2013 operating-system concurrency

Answer key 

36.3

Context Switch (1)

36.3.1 Context Switch: GATE CSE 2000 | Question: 1.20, ISRO2008-47



Which of the following need not necessarily be saved on a context switch between processes?

- A. General purpose registers
- B. Translation look-aside buffer
- C. Program counter
- D. All of the above

gatecse-2000 operating-system easy isro2008 context-switch

Answer key 

36.4

Critical Section (1)

36.4.1 Critical Section: ISRO CSE 2017 | Question: 68



A critical region

- A. is a piece of code which only one process executes at a time
- B. is a region prone to deadlock
- C. is a piece of code which only a finite number of processes execute
- D. is found only in windows NT operating system

isro2017 operating-system critical-section process-synchronization

Answer key 

36.5

Deadlock Prevention Avoidance Detection (4)

36.5.1 Deadlock Prevention Avoidance Detection: ISRO CSE 2009 | Question: 77



When a process is rolled back as a result of deadlock the difficulty which arises is

- A. Starvation
- B. System throughput

C. Low device utilization

D. Cycle stealing

isro2009 deadlock-prevention-avoidance-detection

Answer key

36.5.2 Deadlock Prevention Avoidance Detection: ISRO CSE 2018 | Question: 21



Consider a system having m resources of the same type. These resources are shared by 3 processes A, B, C , which have peak time demands of 3, 4, 6 respectively. The minimum value of m that ensures that deadlock will never occur is:

a. 11

b. 12

c. 13

d. 14

isro2018 deadlock-prevention-avoidance-detection operating-system

Answer key

36.5.3 Deadlock Prevention Avoidance Detection: ISRO CSE 2020 | Question: 29



An aid to determine the deadlock occurrence is

A. resource allocation graph

B. starvation graph

C. inversion graph

D. none of the above

isro-2020 operating-system deadlock-prevention-avoidance-detection easy

Answer key

36.5.4 Deadlock Prevention Avoidance Detection: ISRO-DEC2017-64



Which of the following is not true with respect to deadlock prevention and deadlock avoidance schemes?

A. In deadlock prevention, the request for resources is always granted if resulting state is safe.

B. In deadlock avoidance, the request for resources is always granted. if the resulting state is safe.

C. Deadlock avoidance requires knowledge of resource requirements a priori.

D. Deadlock prevention is more restrictive than deadlock avoidance.

isrodec2017 deadlock-prevention-avoidance-detection operating-system

Answer key

36.6

Disk (11)

36.6.1 Disk: GATE CSE 1999 | Question: 2-18, ISRO2008-46



Raid configurations of the disks are used to provide

A. Fault-tolerance

B. High speed

C. High data density

D. (A) & (B)

gate1999 operating-system disk easy isro2008

Answer key

36.6.2 Disk: GATE CSE 2003 | Question: 25, ISRO2009-12



Using a larger block size in a fixed block size file system leads to

A. better disk throughput but poorer disk space utilization

B. better disk throughput and better disk space utilization

C. poorer disk throughput but better disk space utilization

D. poorer disk throughput and poorer disk space utilization

Answer key **36.6.3 Disk: GATE CSE 2007 | Question: 11, ISRO2009-36, ISRO2016-21**

Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

- | | |
|-----------------------|-----------------------|
| A. 256 Mbyte, 19 bits | B. 256 Mbyte, 28 bits |
| C. 512 Mbyte, 20 bits | D. 64 Gbyte, 28 bits |

Answer key **36.6.4 Disk: GATE IT 2007 | Question: 44, ISRO2015-34**

A hard disk system has the following parameters :

- Number of tracks = 500
- Number of sectors/track = 100
- Number of bytes /sector = 500
- Time taken by the head to move from one track to adjacent track = 1 ms
- Rotation speed = 600 rpm.

What is the average time taken for transferring 250 bytes from the disk ?

- | | | | |
|-------------|-------------|-----------|-----------|
| A. 300.5 ms | B. 255.5 ms | C. 255 ms | D. 300 ms |
|-------------|-------------|-----------|-----------|

Answer key **36.6.5 Disk: ISRO CSE 2007 | Question: 39**

Disk requests are received by a disk drive for cylinder 5, 25, 18, 3, 39, 8 and 35 in that order. A seek takes 5 msec per cylinder moved. How much seek time is needed to serve these requests for a Shortest Seek First (SSF) algorithm? Assume that the arm is at cylinder 20 when the last of these requests is made with none of the requests yet served

- | | | | |
|-------------|-------------|-------------|-------------|
| A. 125 msec | B. 295 msec | C. 575 msec | D. 750 msec |
|-------------|-------------|-------------|-------------|

Answer key **36.6.6 Disk: ISRO CSE 2007 | Question: 56**

Which of the following RAID level provides the highest Data Transfer Rate (Read/Write)

- | | | | |
|-----------|-----------|-----------|-----------|
| A. RAID 1 | B. RAID 3 | C. RAID 4 | D. RAID 5 |
|-----------|-----------|-----------|-----------|

Answer key **36.6.7 Disk: ISRO CSE 2008 | Question: 49**

The total time to prepare a disk drive mechanism for a block of data to be read from it is

- | | |
|---------------------------|----------------------|
| A. seek time | B. latency |
| C. latency plus seek time | D. transmission time |

Answer key **36.6.8 Disk: ISRO CSE 2009 | Question: 36**

Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stores in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively

- A. 256 Mbyte, 19 bits
 C. 512 Mbyte, 20 bits
 B. 256 Mbute, 28 bits
 D. 64 Gbyte, 28 bits

Answer key **36.6.9 Disk: ISRO CSE 2011 | Question: 38**

A fast wide SCSI-II disk drive spins at 7200 RPM, has a sector size of 512 bytes, and holds 160 sectors per track. Estimate the sustained transfer rate of this drive

- A. 576000 Kilobytes / sec
 C. 4800 Kilobytes / sec
 B. 9600 Kilobytes / sec
 D. 19200 Kilobytes / sec

Answer key **36.6.10 Disk: ISRO CSE 2018 | Question: 23**

Disk requests come to a disk driver for cylinders in the order 10, 22, 20, 2, 40, 6 and 38 at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms/cylinder. The total seek time, if the disk arm scheduling algorithms is first-come-first-served is

- a. 360
 b. 850
 c. 900
 d. None of the above

Answer key **36.6.11 Disk: ISRO CSE 2018 | Question: 50**

A particular disk unit uses a bit string to record the occupancy or vacancy of its tracks, with 0 denoting vacant and 1 for occupied. A 32-bit segment of this string has hexadecimal value D4FE2003. The percentage of occupied tracks for the corresponding part of the disk, to the nearest percentage is:

- a. 12
 b. 25
 c. 38
 d. 44

Answer key **36.7****Disk Scheduling (3)****36.7.1 Disk Scheduling: ISRO CSE 2014 | Question: 14**

There are 200 tracks on a disc platter and the pending requests have come in the order - 36, 69, 167, 76, 42, 51, 126, 12 and 199. Assume the arm is located at the 100th track and moving towards track 200. If sequence of disc access is 126, 167, 199, 12, 36, 42, 51, 69 and 76 then which disc access scheduling policy is used?

- A. Elevator
 C. C-SCAN
 B. Shortest seek-time first
 D. First Come First Served

Answer key **36.7.2 Disk Scheduling: ISRO CSE 2017 | Question: 66**

Consider the disk system with 100 cylinders. The request to access the cylinders occur in the following sequence.

4, 37, 10, 7, 19, 73, 2, 15, 6, 20

Assuming the head is currently at cylinder 50 what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first algorithm is used.

- A. 95 ms B. 119 ms C. 233 ms D. 276 ms

Answer key **36.7.3 Disk Scheduling: ISRO-DEC2017-69**

Consider a disk sequence with 100 cylinders. The request to access the cylinder occur in the following sequence :

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 2 ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

- A. 190 B. 238 C. 233 D. 276

Answer key **36.8 Fifo (1)****36.8.1 Fifo: ISRO-DEC2017-62**

A system uses *FIFO* policy for page replacement. It has 4 page frames with no pages loaded, to begin with. The system first accesses 50 distinct pages in some order and then accesses the same 50 pages in reverse order. How many page faults will occur?

- A. 96 B. 100 C. 97 D. 92

Answer key **36.9 File System (1)****36.9.1 File System: ISRO CSE 2014 | Question: 66**

Consider a system where each file is associated with a 16 bit number. For each file, each user should have the read and write capability. How much memory is needed to store each user's access data?

- A. 16 KB B. 32 KB C. 64 KB D. 128 KB

Answer key **36.10 Fork System Call (4)**

36.10.1 Fork System Call: ISRO CSE 2008 | Question: 63



Fork is

- A. the creation of a new job
- B. the dispatching of a task
- C. increasing the priority of a task
- D. the creation of a new process

isro2008 operating-system fork-system-call easy

Answer key

36.10.2 Fork System Call: ISRO CSE 2017 | Question: 51



What is the output of the following program?

```
main()
{
    int a = 10;
    if(fork() == 0)
        a++;
    printf("%d\n",a);
}
```

- A. 10 and 11
- B. 10
- C. 11
- D. 11 and 11

isro2017 operating-system fork-system-call

Answer key

36.10.3 Fork System Call: ISRO CSE 2018 | Question: 4



The following C program:

```
{
    fork(); fork(); printf("yes");
}
```

If we execute this code segment, how many times the string yes will be printed?

- a. Only once
- b. 2 times
- c. 4 times
- d. 8 times

isro2018 fork-system-call operating-system easy

Answer key

36.10.4 Fork System Call: ISRO2015-37



Consider the following program.

```
main()
{
    fork();
    fork();
    fork();
}
```

How many new processes will be created?

- A. 8
- B. 6
- C. 7
- D. 5

isro2015 operating-system fork-system-call easy

Answer key

36.11.1 Fragmentation: ISRO CSE 2018 | Question: 22



A computer has $1000K$ of main memory. The jobs arrive and finish in the following sequence.

Job 1 requiring $200K$ arrives

Job 2 requiring $350K$ arrives

Job 3 requiring $300K$ arrives

Job 1 finishes

Job 4 requiring $120K$ arrives

Job 5 requiring $150K$ arrives

Job 6 requiring $80K$ arrives

Among best fit and first fit, which performs better for this sequence?

- a. First fit
- b. Best fit
- c. Both perform the same
- d. None of the above

isro2018 memory-management fragmentation operating-system

Answer key

36.12

IO Devices (1)

36.12.1 IO Devices: ISRO-2013-51



In case of a DVD, the speed of data transfer is mentioned in multiples of?

- A. 150 KB/s
- B. 1.38 MB/s
- C. 300 KB/s
- D. 2.40 MB/s

isro2013 io-devices non-gatecse out-of-syllabus-now

Answer key

36.13

IO Handling (3)

36.13.1 IO Handling: GATE CSE 1996 | Question: 1.20, ISRO2008-56



Which of the following is an example of spooled device?

- A. A line printer used to print the output of a number of jobs
- B. A terminal used to enter input data to a running program
- C. A secondary storage device in a virtual memory system
- D. A graphic display device

gate1996 operating-system io-handling normal isro2008

Answer key

36.13.2 IO Handling: GATE IT 2004 | Question: 11, ISRO2011-33



What is the bit rate of a video terminal unit with 80 characters/line, 8 bits/character and horizontal sweep time of $100\mu s$ (including $20\mu s$ of retrace time)?

- A. 8 Mbps
- B. 6.4 Mbps
- C. 0.8 Mbps
- D. 0.64 Mbps

gateit-2004 operating-system io-handling easy isro2011

Answer key

36.13.3 IO Handling: ISRO-2013-52



Suppose we have variable logical records of lengths of 5 bytes, 10 bytes and 25 bytes while the physical block size in disk is 15 bytes. What is the maximum and minimum fragmentation seen in bytes?

- A. 25 and 5 B. 15 and 5 C. 15 and 0 D. 10 and 5

isro2013 operating-system io-handling disk

Answer key

36.14

Macros (1)

36.14.1 Macros: ISRO CSE 2018 | Question: 67



In multi-programmed systems, it is advantageous if some programs such as editors and compilers can be shared by several users.

Which of the following must be true of multi-programmed systems in order that a single copy of a program can be shared by several users?

- I. The program is a macro
- II. The program is recursive
- III. The program is reentrant

- a. I only b. II only c. III only d. I, II and III

isro2018 operating-system macros

Answer key

36.15

Memory Management (6)

36.15.1 Memory Management: ISRO CSE 2018 | Question: 61



The Operating System of a computer may periodically collect all the free memory space to form contiguous block of free space. This is called:

- a. Concatenation b. Garbage Collection
c. Collision d. Dynamic Memory Allocation

isro2018 memory-management operating-system

Answer key

36.15.2 Memory Management: ISRO CSE 2020 | Question: 25



What is compaction refers to

- A. a technique for overcoming internal fragmentation
- B. a paging technique
- C. a technique for overcoming external fragmentation
- D. a technique for compressing the data

isro-2020 operating-system memory-management easy

Answer key

36.15.3 Memory Management: ISRO CSE 2023 | Question: 2



Which of the following heap memory E allocation strategies is likely to exploit spatial locality of memory accesses in a program, the most?

A. Best Fit

B. First Fit

C. Next Fit

D. Last Fit

isro-cse-2023 operating-system memory-management

Answer key

36.15.4 Memory Management: ISRO CSE 2023 | Question: 76



Any attempt by a process to access memory allocated to OS or process of other user results into

A. Trap to OS

B. Context Switching

C. Page fault

D. Invocation to Scheduler Despatch

isro-cse-2023 operating-system memory-management

Answer key

36.15.5 Memory Management: ISRO-2013-56



In a 64-bit machine, with 2 GB RAM, and 8 KB page size, how many entries will be there in the page table if its is inverted?

A. 2^{18}

B. 2^{20}

C. 2^{33}

D. 2^{51}

isro2013 operating-system memory-management paging

Answer key

36.15.6 Memory Management: ISRO2013-55



Consider a logical address space of 8 pages of 1024 words each, mapped onto a physical memory of 32 frames. How many bits are there in the physical address and logical address respectively?

A. 5,3

B. 10,10

C. 15,13

D. 15,15

isro2013 operating-system memory-management

Answer key

36.16

Multilevel Paging (1)

36.16.1 Multilevel Paging: ISRO CSE 2011 | Question: 49



Consider a 32-bit machine where four-level paging scheme is used. If the hit ratio to TLB is 98%, and it takes 20 nanosecond to search the TLB and 100 nanoseconds to access the main memory what is effective memory access time in nanoseconds?

A. 126

B. 128

C. 122

D. 120

isro2011 operating-system multilevel-paging

Answer key

36.17

Mutual Exclusion (2)

36.17.1 Mutual Exclusion: ISRO CSE 2017 | Question: 56



Mutual exclusion problem occurs

A. between two disjoint processes that do not interact

- B. among processes that share resources
- C. among processes that do not use the same resource
- D. between two processes that uses different resources of different machine

isro2017 operating-system process-synchronization mutual-exclusion

Answer key

36.17.2 Mutual Exclusion: ISRO CSE 2020 | Question: 57



The hardware implementation which provides mutual exclusion is

- A. Semaphores
- B. Test and set instructions
- C. Both options
- D. None of the options

isro-2020 operating-system semaphore mutual-exclusion normal

Answer key

36.18 Page Replacement (13)

36.18.1 Page Replacement: GATE CSE 1997 | Question: 3.10, ISRO2008-57, ISRO2015-64



Dirty bit for a page in a page table

- A. helps avoid unnecessary writes on a paging device
- B. helps maintain LRU information
- C. allows only read on a page
- D. None of the above

gate1997 operating-system page-replacement easy isro2008 isro2015

Answer key

36.18.2 Page Replacement: GATE CSE 2004 | Question: 21, ISRO2007-44



The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by

- A. the instruction set architecture
- B. page size
- C. number of processes in memory
- D. physical memory size

gatecse-2004 operating-system virtual-memory page-replacement normal isro2007

Answer key

36.18.3 Page Replacement: GATE CSE 2005 | Question: 22, ISRO2015-36



Increasing the RAM of a computer typically improves performance because:

- A. Virtual Memory increases
- B. Larger RAMs are faster
- C. Fewer page faults occur
- D. Fewer segmentation faults occur

gatecse-2005 operating-system page-replacement easy isro2015

Answer key

36.18.4 Page Replacement: GATE CSE 2009 | Question: 9, ISRO2016-52



In which one of the following page replacement policies, Belady's anomaly may occur?

- A. FIFO
- B. Optimal
- C. LRU
- D. MRU

gatecse-2009 operating-system page-replacement normal isro2016

Answer key

36.18.5 Page Replacement: ISRO CSE 2007 | Question: 10



The term 'aging' refers to

- A. booting up the priority of the process in multi-level of queue without feedback.
- B. gradually increasing the priority of jobs that wait in the system for a long time to remedy infinite blocking
- C. keeping track of the following a page has been in memory for the purpose of LRU replacement
- D. letting job reside in memory for a certain amount of time so that the number of pages required can be estimated accurately.

isro2007 operating-system memory-management page-replacement

Answer key

36.18.6 Page Replacement: ISRO CSE 2008 | Question: 67



The page replacement algorithm which gives the lowest page fault rate is

- A. LRU
- B. FIFO
- C. Optimal page replacement
- D. Second chance algorithm

isro2008 operating-system page-replacement

Answer key

36.18.7 Page Replacement: ISRO CSE 2011 | Question: 73



Belady's anomaly means

- A. Page fault rate is constant even on increasing the number of allocated frames
- B. Page fault rate may increase on increasing the number of allocated frames
- C. Page fault rate may increase on decreasing the number of allocated frames
- D. Page fault rate may decrease on increasing the number of allocated frames

isro2011 operating-system page-replacement page-fault

Answer key

36.18.8 Page Replacement: ISRO CSE 2016 | Question: 48



Determine the number of page faults when references to pages occur in the following order:

1, 2, 4, 5, 2, 1, 2, 4

Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page one having brought earlier than page 2. (LRU page replacement algorithm is used)

- A. 3
- B. 5
- C. 4
- D. None of these

page-replacement page-fault isro2016 operating-system

Answer key

36.18.9 Page Replacement: ISRO CSE 2016 | Question: 49



Working Set (t, k) at an instant of time t is

- A. the set of k future references that the OS will make
- B. the set of future references that the OS will make in next t unit of time
- C. the set of k references with high frequency
- D. the k set of pages that have been referenced in the last t time units

Answer key

36.18.10 Page Replacement: ISRO CSE 2018 | Question: 20



Determine the number of page faults when references to pages occur in the order 1,2,4,5,2,1,2,4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 brought earlier than page 2. (assume LRU i.e., Least-Recently-Used algorithm is applied)

- A. 3 B. 4 C. 5 D. None of the above

isro2018 operating-system page-replacement

Answer key

36.18.11 Page Replacement: ISRO CSE 2020 | Question: 30



Consider the following page reference string.

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

What are the minimum number of frames required to get a single page fault for the above sequence assuming LRU replacement strategy?

- A. 7 B. 4 C. 6 D. 5

isro-2020 operating-system memory-management page-replacement page-fault normal

Answer key

36.18.12 Page Replacement: ISRO-2013-65



Consider the list of page references in the time line as below:

9 6 2 3 4 4 4 4 3 4 4 2 5 8 6 8 5 5 3 2 3 3 9 6 2 7

What is the working set at the penultimate page reference if Δ is 5?

- A. {8, 5, 3, 2, 9, 6} B. {4, 3, 6, 2, 5}
C. {3, 9, 6, 2, 7} D. {3, 9, 6, 2}

isro2013 operating-system memory-management page-replacement

Answer key

36.18.13 Page Replacement: ISRO-DEC2017-58



A virtual memory system uses *FIFO* page replacement policy and allocates a fixed number of frames to the process. Consider the following statements

Increasing the number of page frames
M: allocated to a process sometimes increases the page fault rate.

N: Some programs do not exhibit locality of reference.

Which one of the following is true?

- A. Both M and N are true and N is the reason for M .
B. Both M and N are true but N is not the reason for M .
C. Both M and N are false.
D. M is false, but N is true.

isrodec2017 operating-system page-replacement

Answer key 

36.19

Pipes (1)

36.19.1 Pipes: ISRO CSE 2018 | Question: 13



The difference between a named pipe and a regular file in Unix is that

- a. Unlike a regular file, named pipe is a special file
- b. The data in a pipe is transient, unlike the content of a regular file
- c. Pipes forbid random accessing, while regular files do allow this.
- d. All of the above

isro2018 operating-system pipes non-gatecse

Answer key 

36.20

Process (8)

36.20.1 Process: ISRO CSE 2007 | Question: 41



A task in a blocked state

- | | |
|---|--|
| A. is executable | B. is running |
| C. must still be placed in the run queues | D. is waiting for some temporarily unavailable resources |

isro2007 operating-system process

Answer key 

36.20.2 Process: ISRO CSE 2008 | Question: 64



Which of the following need not necessarily be saved on a Context Switch between processes?

- | | |
|------------------------------|----------------------------------|
| A. General purpose registers | B. Translation look-aside buffer |
| C. Program counter | D. Stack pointer |

isro2008 operating-system process

Answer key 

36.20.3 Process: ISRO CSE 2009 | Question: 75



Special software to create a job queue is called a

- | | | | |
|-----------|------------|----------------|-------------------|
| A. Driver | B. Spooler | C. Interpreter | D. Linkage editor |
|-----------|------------|----------------|-------------------|

isro2009 operating-system process

Answer key 

36.20.4 Process: ISRO CSE 2009 | Question: 76



Process is

- | | |
|--|------------------------------|
| A. A program in high level language kept on disk | B. Contents of main memory |
| C. A program in execution | D. A job in secondary memory |

isro2009 operating-system process

Answer key 

36.20.5 Process: ISRO CSE 2009 | Question: 9



Which is the correct definition of a valid process transition in an operating system?

- A. Wake up: ready $\rightarrow$ running
- B. Dispatch: ready $\rightarrow$ running
- C. Block: ready $\rightarrow$ running
- D. Timer runout: $\rightarrow$ ready $\rightarrow$ running

isro2009 operating-system process

Answer key

36.20.6 Process: ISRO CSE 2011 | Question: 55



There are three processes in the ready queue. When the currently running process requests for I/O how many process switches take place?

- A. 1
- B. 2
- C. 3
- D. 4

isro2011 operating-system process

Answer key

36.20.7 Process: ISRO-2013-60



The state of a process after it encounters an I/O instruction is?

- A. Ready
- B. Blocked
- C. Idle
- D. Running

isro2013 operating-system process

Answer key

36.20.8 Process: ISRO-DEC2017-60



A process executes the following code

```
for(i=0;i<n;i++)  
    fork();
```

The total number of child processes created is

- A. n^2
- B. $2^{n+1} - 1$
- C. 2^n
- D. $2^n - 1$

isrodec2017 process fork-system-call operating-system

Answer key

36.21 Process Scheduling (23)

36.21.1 Process Scheduling: GATE CSE 1996 | Question: 2.20, ISRO2008-15



Four jobs to be executed on a single processor system arrive at time 0 in the order A, B, C, D . Their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is

- A. 10
- B. 4
- C. 8
- D. 9

gate1996 operating-system process-scheduling normal isro2008

Answer key

36.21.2 Process Scheduling: GATE CSE 2006 | Question: 06, ISRO2009-14



Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at

time zero and at the end.

- A. 1 B. 2 C. 3 D. 4

gatecse-2006 operating-system process-scheduling normal isro2009

Answer key 

36.21.3 Process Scheduling: ISRO CSE 2007 | Question: 11, GATE CSE 2001 | Question: 1.19

Consider a set of n tasks with known runtimes $r_1, r_2, \dots, r_n$ to be run on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?



- A. Round Robin B. Shortest job first
C. Highest response ratio next D. first come first served

isro2007 operating-system process-scheduling gatecse-2001

Answer key 

36.21.4 Process Scheduling: ISRO CSE 2007 | Question: 12

Consider a job scheduling problem with 4 jobs J_1, J_2, J_3 and J_4 with corresponding deadlines: $(d_1, d_2, d_3, d_4) = (4, 2, 4, 2)$. Which of the following is not a feasible schedule without violating any job schedule?



- A. J_2, J_4, J_1, J_3 B. J_4, J_1, J_2, J_3
C. J_4, J_2, J_1, J_3 D. J_4, J_2, J_3, J_1

isro2007 operating-system process-scheduling

Answer key 

36.21.5 Process Scheduling: ISRO CSE 2007 | Question: 17

Round Robin schedule is essentially the pre-emptive version of



- A. FIFO B. Shortest job first
C. Shortest remaining time D. Longest remaining time

isro2007 operating-system process-scheduling

Answer key 

36.21.6 Process Scheduling: ISRO CSE 2007 | Question: 43

On a system using non-preemptive scheduling, processes with expected run times of 5, 18, 9 and 12 are in the ready queue. In what order should they be run to minimize wait time?



- A. 5, 12, 9, 18 B. 5, 9, 12, 18 C. 12, 18, 9, 5 D. 9, 12, 18, 5

isro2007 operating-system process-scheduling

Answer key 

36.21.7 Process Scheduling: ISRO CSE 2007 | Question: 64, ISRO CSE 2008 | Question: 50

Feedback queues



- A. are very simple to implement
B. dispatch tasks according to execution characteristics
C. are used to favour real time tasks
D. require manual intervention to implement properly

isro2007 operating-system isro2008 process-scheduling

Answer key**36.21.8 Process Scheduling: ISRO CSE 2008 | Question: 51**

With Round-Robin CPU scheduling in a time shared system

- A. using very large time slices (quantas) degenerates into First-Come First served (FCFS) algorithm.
- B. using extremely small time slices improves performance
- C. using very small time slices degenerates into Last-In First-Out (LIFO) algorithm.
- D. using medium sized times slices leads to shortest Request time First (SRTF) algorithm

isro2008 operating-system process-scheduling

Answer key**36.21.9 Process Scheduling: ISRO CSE 2008 | Question: 66, ISRO CSE 2009 | Question: 15**

The performance of Round Robin algorithm depends heavily on

- A. size of the process
- B. the I/O bursts of the process
- C. the CPU bursts of the process
- D. the size of the time quantum

isro2008 isro2009 operating-system process-scheduling

Answer key**36.21.10 Process Scheduling: ISRO CSE 2009 | Question: 10**

The correct matching of the following pairs is

A.	Disk check	i.	Round robin
B.	Batch processing	ii.	Scan
C.	Time sharing	iii.	LIFO
D.	Stack operation	iv.	FIFO

- 1. a-iii, b-iv, c-ii, d-i
- 2. a-iv, b-iii, c-ii, d-i
- 3. a-iii, b-iv, c-i, d-ii
- 4. a-ii, b-iv, c-i, d-iii

isro2009 operating-system process-scheduling

Answer key**36.21.11 Process Scheduling: ISRO CSE 2009 | Question: 17**

Consider a set of 5 processes whose arrival time, CPU time needed and the priority are given below:

Process Priority	Arrival Time ms)	(in CPU Time Needed	Priority
P1	0	10	5
P2	0	5	2
P3	2	3	1
P4	5	20	4
P5	10	2	3

(smaller the number, higher the priority)

If the CPU scheduling policy is priority scheduling without pre-emption, the average waiting time will be

- A. 12.8 ms B. 11.8 ms C. 10.8 ms D. 09.8 ms

isro2009 operating-system process-scheduling

Answer key 

36.21.12 Process Scheduling: ISRO CSE 2011 | Question: 4



The following table shows the processes in the ready queue and time required for each process for completing its job.

Process Time

P_1	10
P_2	5
P_3	20
P_4	8
P_5	15

If round-robin scheduling with 5 ms is used what is the average waiting time of the processes in the queue?

- A. 27 ms B. 26.2 ms C. 27.5 ms D. 27.2 ms

isro2011 operating-system process-scheduling

Answer key 

36.21.13 Process Scheduling: ISRO CSE 2014 | Question: 78



Which of the following is not an optimization criterion in the design of a CPU scheduling algorithm?

- A. Minimum CPU utilization B. Maximum throughput
C. Minimum turnaround time D. Minimum waiting time

isro2014 operating-system process-scheduling

Answer key 

36.21.14 Process Scheduling: ISRO CSE 2016 | Question: 51



For the real time operating system, which of the following is the most suitable scheduling scheme?

- A. Round robin B. First come first serve
C. Pre-emptive D. Random scheduling

isro2016 operating-system process-scheduling

Answer key 

36.21.15 Process Scheduling: ISRO CSE 2020 | Question: 27



Which of the following algorithms defines time quantum?

- A. shortest job scheduling algorithm B. round robin scheduling algorithm
C. priority scheduling algorithm D. multilevel queue scheduling algorithm

isro-2020 operating-system process-scheduling easy

Answer key 

36.21.16 Process Scheduling: ISRO CSE 2020 | Question: 31

Three CPU-bound tasks, with execution times of 15, 12 and 5 time units respectively arrive at times 0, t and 8, respectively. If the operating system implements a shortest remaining time first scheduling algorithm, what should be the value of t to have 4 context switches? Ignore the context switches at time 0 and at the end.

- A. $0 < t < 3$ B. $t = 0$ C. $t \leq 3$ D. $3 < t < 8$

isro-2020 operating-system process-scheduling normal

Answer key

36.21.17 Process Scheduling: ISRO CSE 2023 | Question: 54

	Process Scheduler Type		Activity
(i)	Long Term Scheduler	(P)	executes much faster to reduce wastage of CPU time
(ii)	Medium term scheduler	(Q)	controls degree of multi programming
(iii)	Short Term scheduler	(R)	is associated with swapping

with respect to Process Schedulers of an operating system mentioned above in the table, which of the following statements is correct?

- a. (i) – (R), (ii) – (P), (iii) – (Q)
 b. (i) – (Q), (ii) – (R), (iii) – (P)
 c. (i) – (P), (ii) – (R), (iii) – (Q)
 d. (i) – (R), (ii) – (Q), (iii) – (P)

isro-cse-2023 operating-system process-scheduling

Answer key

36.21.18 Process Scheduling: ISRO-2013-50

Consider the following set of processes, with arrival times and the required CPU-burst times given in milliseconds.

Process	Arrival time	Burst Time
P1	0	4
P2	2	2
P3	3	1

What is the sequence in which the processes are completed? Assume round robin scheduling with a time quantum of 2 milliseconds?

- A. P1, P2, P3 B. P2, P1, P3 C. P3, P2, P1 D. P2, P3, P1

isro2013 process-scheduling round-robin-scheduling

Answer key

36.21.19 Process Scheduling: ISRO-2013-59

A starvation free job scheduling policy guarantees that no job indefinitely waits for a service. Which of the following job scheduling policies is starvation free?

- A. Priority queuing B. Shortest job first
 C. Youngest job first D. Round robin

isro2013 operating-system process-scheduling

Answer key

36.21.20 Process Scheduling: ISRO-2013-70



Which of the following strategy is employed for overcoming the priority inversion problem?

- A. Temporarily raise the priority of lower priority level process
- B. Have a fixed priority level scheme.
- C. Implement Kernel pre-emption scheme.
- D. Allow lower priority process to complete its job.

isro2013 operating-system process-scheduling

Answer key

36.21.21 Process Scheduling: ISRO-DEC2017-61



Consider the following scheduling:

A.	Gang scheduling	s.	Guaranteed scheduling
B.	Rate monotonic scheduling	t.	Thread scheduling
C.	Fair share scheduling	u.	Real-time scheduling

Matching the table in the order A, B, C gives

- A. t,u,s
- B. s,t,u
- C. u,s,t
- D. u,t,s

isrodec2017 process-scheduling operating-system

Answer key

36.21.22 Process Scheduling: ISRO2015-33



Suppose a system contains n processes and system uses the round-robin algorithm for CPU scheduling then which data structure is best suited ready queue of the process

- A. stack
- B. queue
- C. circular queue
- D. tree

isro2015 operating-system process-scheduling

Answer key

36.21.23 Process Scheduling: ISRO2015-38



Suppose two jobs, each of which needs 10 minutes of CPU time, start simultaneously. Assume 50% I/O wait time. How long will it take for both to complete, if they run sequentially?

- A. 10
- B. 20
- C. 30
- D. 40

process-scheduling isro2015

Answer key

36.22

Process Synchronization (7)

36.22.1 Process Synchronization: GATE CSE 1996 | Question: 1.19, ISRO2008-61



A critical section is a program segment

- A. which should run in a certain amount of time
- B. which avoids deadlocks
- C. where shared resources are accessed
- D. which must be enclosed by a pair of semaphore operations, P and V

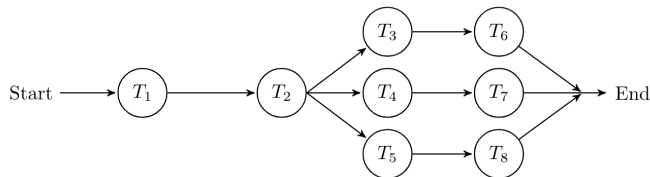
gate1996 operating-system process-synchronization easy isro2008

Answer key

36.22.2 Process Synchronization: ISRO CSE 2011 | Question: 10



Below is the precedence graph for a set of tasks to be executed on a parallel processing system S .



What is the efficiency of this precedence graph on S if each of the tasks $T_1, \dots, T_8$ takes the same time and the system S has five processors?

- A. 25%
- B. 40%
- C. 50%
- D. 90%

isro2011 operating-system process-synchronization

Answer key

36.22.3 Process Synchronization: ISRO CSE 2011 | Question: 77



In a system using single processor, a new process arrives at the rate of six processes per minute and each such process requires seven seconds of service time. What is the CPU utilization?

- A. 70%
- B. 30%
- C. 60%
- D. 64%

isro2011 operating-system process-synchronization

Answer key

36.22.4 Process Synchronization: ISRO CSE 2018 | Question: 67



Procedures $P1$ and $P2$ have a producer-consumer relationship, communicating by the use of a set of shared buffers.

$P1$: repeat

Obtain an empty buffer

Fill it

Return a full buffer

forever

$P2$: repeat

Obtain a full buffer

Empty it

Return an empty buffer

forever

Increasing the number of buffers is likely to do which of the following?

- I. Increase the rate at which requests are satisfied (throughput)
- II. Decrease the likelihood of deadlock
- III. Increase the ease of achieving a correct implementation

a. III only b. II only c. I only d. II and III only

isro2018 process-synchronization operating-system

Answer key 

36.22.5 Process Synchronization: ISRO CSE 2020 | Question: 26



The operating system and the other processes are protected from being modified by an already running process because

- A. they run at different time instants and not in parallel
- B. they are in different logical addresses
- C. they use a protection algorithm in the scheduler
- D. every address generated by the CPU is being checked against the relocation and limit parameters

isro-2020 operating-system process-synchronization easy

Answer key 

36.22.6 Process Synchronization: ISRO CSE 2020 | Question: 28



Dispatch latency is defined as

- A. the speed of dispatching a process from running to the ready state
- B. the time of dispatching a process from running to ready state and keeping the CPU idle
- C. the time to stop one process and start running another one
- D. none of these

isro-2020 operating-system process-synchronization easy

Answer key 

36.22.7 Process Synchronization: ISRO CSE 2023 | Question: 75



In a producer-consumer scenario also known as Bounded-Buffer Problem, what would be the most appropriate synchronization primitive to ensure that the consumer waits when the buffer is empty?

- A. Spinlock B. Mutex lock C. Semaphore D. Monitors

isro-cse-2023 operating-system process-synchronization

Answer key 

36.23

Process and Threads (1)

36.23.1 Process and Threads: ISRO-DEC2017-63



Which of the following is false?

- A. User level threads are not scheduled by the kernel.
- B. Context switching between user-level threads is faster than context switching between kernel-level threads.

- C. When a user thread is blocked all other threads of its processes are blocked.
D. Kernel-level threads cannot utilize multiprocessor systems by splitting threads on different processors or cores.

isrodec2017 operating-system process-and-threads threads

Answer key 

36.24 Realtime Systems (1)

36.24.1 Realtime Systems: ISRO CSE 2017 | Question: 73



Which of the following statement is true?

- A. Hard real time OS has less jitter than soft real time OS
B. Hard real time OS has more jitter than soft real time OS
C. Hard real time OS has equal jitter as soft real time OS
D. None of the above

isro2017 operating-system realtime-systems

Answer key 

36.25 Resource Allocation (10)

36.25.1 Resource Allocation: ISRO CSE 2007 | Question: 40



Consider a system having 'm' resources of the same type. The resources are shared by 3 processes A, B, C, which have peak time demands of 3, 4, 6 respectively. The minimum value of 'm' that ensures that deadlock will never occur is

- A. 11 B. 12 C. 13 D. 14

isro2007 operating-system resource-allocation

Answer key 

36.25.2 Resource Allocation: ISRO CSE 2008 | Question: 62



In which of the following four necessary conditions for deadlock processes claim exclusive control of the resources they require?

- A. no preemption B. mutual exclusion
C. circular wait D. hold and wait

isro2008 operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key 

36.25.3 Resource Allocation: ISRO CSE 2009 | Question: 16



Consider a system having "n" resources of same type. These resources are shared by 3 processes, A, B, C. These have peak demands of 3, 4, and 6 respectively. For what value of "n" deadlock won't occur

- A. 15 B. 9 C. 10 D. 13

isro2009 operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key 

36.25.4 Resource Allocation: ISRO CSE 2011 | Question: 60



A total of 9 units of a resource type available, and given the safe state shown below, which of the following sequence will be a safe state?

Process Used Max

P_1	2	7
P_2	1	6
P_3	2	5
P_4	1	4

- A. $\langle P_4, P_1, P_3, P_2 \rangle$
B. $\langle P_4, P_2, P_1, P_3 \rangle$
C. $\langle P_4, P_2, P_3, P_1 \rangle$
D. $\langle P_3, P_1, P_2, P_4 \rangle$

isro2011 operating-system resource-allocation

Answer key

36.25.5 Resource Allocation: ISRO CSE 2014 | Question: 68



What is the minimum number of resources required to ensure that deadlock will never occur, if there are currently three processes P_1, P_2 and P_3 running in a system whose maximum demand for the resources of same type are 3, 4, and 5 respectively.

- A. 3
B. 7
C. 9
D. 10

isro2014 operating-system resource-allocation

Answer key

36.25.6 Resource Allocation: ISRO CSE 2016 | Question: 46



With single resource, deadlock occurs

- A. if there are more than two processes competing for that resources
B. if there are only two processes competing for that resources
C. if there is a single process competing for that resources
D. none of these

operating-system resource-allocation isro2016

Answer key

36.25.7 Resource Allocation: ISRO CSE 2016 | Question: 47



A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units, then

- A. Deadlock can never occur
B. Deadlock may occur
C. Deadlock has to occur
D. None of these

operating-system resource-allocation isro2016

Answer key

36.25.8 Resource Allocation: ISRO-2013-57



Which of the following is not a necessary condition for deadlock?

- A. Mutual exclusion
B. Reentrancy
C. Hold and wait
D. No pre-emption

isro2013 operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

36.25.9 Resource Allocation: ISRO-2013-58



Consider the following process and resource requirement of each process.

Process	Type 1		Type 2	
	Used	Max	Used	Max
P1	1	2	1	3
P2	1	3	1	2
P3	2	4	1	4

Predict the state of this system, assuming that there are a total of 5 instances of resource type 1 and 4 instances of resource type 2.

- A. Can go to safe or unsafe state based on sequence
 B. Safe state
 C. Unsafe state
 D. Deadlock state

isro2013 operating-system resource-allocation

Answer key

36.25.10 Resource Allocation: ISRO2015-32



In a lottery scheduler with 40 tickets, how we will distribute the tickets among 4 processes P_1, P_2, P_3 and P_4 such that each process gets 10%, 5%, 60% and 25% respectively?

	P_1	P_2	P_3	P_4
A	12	4	70	30
B	7	5	20	10
C	4	2	24	10
D	8	5	30	40

isro2015 operating-system resource-allocation

Answer key

36.26

Runtime Environment (1)

36.26.1 Runtime Environment: ISRO-DEC2017-68



Consider the following table :

A.	Activation record	p.	Linking loader
B.	Location counter	q.	Garbage collection
C.	Reference counts	r.	Subroutine call
D.	Address relocation	s.	Assembler

- A. $a - p, b - q, c - r, d - s$
 B. $a - q, b - r, c - s, d - p$
 C. $a - r, b - s, c - q, d - p$
 D. $a - r, b - s, c - p, d - q$

isrodec2017 runtime-environment compiler-design

Answer key

36.27

Segmentation (2)

36.27.1 Segmentation: ISRO CSE 2014 | Question: 18



Consider the following segment table in segmentation scheme :

Segment ID	Base	Limit
0	200	200
1	5000	1210
2	1527	498
3	2500	50

What happens if the logical address requested is – Segment Id 2 and offset 1000?

- A. Fetches the entry at the physical address 2527 for segment Id 2
- B. A trap is generated
- C. Deadlock
- D. Fetches the entry at offset 27 in Segment Id 3

segmentation operating-system isro2014

Answer key

36.27.2 Segmentation: ISRO2015-31



If there are 32 segments, each size 1 k bytes, then the logical address should have

- A. 13 bits
- B. 14 bits
- C. 15 bits
- D. 16 bits

isro2015 operating-system virtual-memory segmentation

Answer key

36.28

Semaphore (6)

36.28.1 Semaphore: GATE CSE 1992 | Question: 02,x, ISRO2015-35



At a particular time of computation, the value of a counting semaphore is 7. Then 20 P operations and 15 V operations were completed on this semaphore. The resulting value of the semaphore is :

- A. 42
- B. 2
- C. 7
- D. 12

gate1992 operating-system semaphore easy isro2015 process-synchronization

Answer key

36.28.2 Semaphore: ISRO CSE 2007 | Question: 42, UGCNET-June2010-II: 37



Semaphores

- A. synchronize critical resources to prevent deadlock
- B. synchronize critical resources to prevent contention
- C. are used to do I/O
- D. are used for memory management

isro2007 operating-system process-synchronization ugcnetcse-june2010-paper2 semaphore

Answer key

36.28.3 Semaphore: ISRO CSE 2016 | Question: 45



At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and x V operations were completed on this semaphore. If the new value of semaphore is 5, x will be

- A. 18 B. 22 C. 15 D. 13

operating-system process-synchronization semaphore isro2016

Answer key

36.28.4 Semaphore: ISRO CSE 2017 | Question: 71



At a particular time the value of counting semaphore is 10. It will become 7 after:

- A. 3 V operations B. 3 P operations
C. 5 V operations and 2 P operations D. 2 V operations and 5 P operations

isro2017 operating-system semaphore

Answer key

36.28.5 Semaphore: ISRO-DEC2017-70



A counting semaphore was initialized to 7. Then 20 P (wait) operations and x V (signal) operations were completed on this semaphore. If the final value of semaphore is 5, then the value x will be

- A. 0 B. 13 C. 18 D. 5

isrodec2017 semaphore easy

Answer key

36.28.6 Semaphore: ISRO2015-30



Semaphores are used to solve the problem of

- i. Race Condition
- ii. Process Synchronization
- iii. Mutual Exclusion
- iv. None of the above

- A. I and II
B. II and III
C. All of the above
D. None of the above

semaphore isro2015 process-synchronization easy

Answer key

36.29

Thrashing (1)

36.29.1 Thrashing: ISRO-2013-54



Which of the following are the likely causes of thrashing?

- A. Page size was very small.
B. There are too many users connected to the system.
C. Least recently used policy is used for page replacement.
D. First in First out policy is used for page replacement.

Answer key 

36.30

Threads (3)

36.30.1 Threads: ISRO CSE 2008 | Question: 60



Overlaying

- A. requires use of a loader
- B. allows larger programs, but requires more effort
- C. is most used on large computers
- D. is transparent to the user

isro2008 operating-system threads

Answer key 

36.30.2 Threads: ISRO CSE 2023 | Question: 11



Which of the following multithreading model is followed in Linux OS?

- A. One User Thread mapping to One Kernel Thread
- B. Many User Threads mapping to One Kernel Thread
- C. One User Thread mapping to Many Kernel Threads
- D. Many User Threads mapping to Many Kernel Threads

isro-cse-2023 operating-system threads

Answer key 

36.30.3 Threads: ISRO CSE 2023 | Question: 12



Purpose of priority inheritance protocol in synchronization mechanisms is

- A. To prevent priority inversion in systems having two or lesser priorities for a resource
- B. To provide mutual exclusion between threads and prevent data corruption
- C. To prevent priority inversion and ensure fairness in resource allocation
- D. To allow multiple threads to access any resource simultaneously

isro-cse-2023 operating-system process-synchronization threads

Answer key 

36.31

Unix (1)

36.31.1 Unix: ISRO CSE 2017 | Question: 72



The Linux command mknod myfifo b 4 16

- A. will create a character device if user is root
- B. will create a named pipe FIFO if user is root
- C. will create a block device if user is root
- D. None of these

isro2017 operating-system unix non-gatecse

Answer key 

36.32

Virtual Memory (10)

Answer key

36.32.6 Virtual Memory: ISRO CSE 2014 | Question: 44



A computer has 16 pages of virtual address space but the size of main memory is only four frames. Initially the memory is empty. A program references the virtual pages in the order 0, 2, 4, 5, 2, 4, 3, 11, 2, 10. How many page faults occur if LRU page replacement algorithm is used?

- A. 3 B. 5 C. 7 D. 8

isro2014 operating-system virtual-memory paging

Answer key

36.32.7 Virtual Memory: ISRO CSE 2014 | Question: 70



Dirty bit is used to indicate which of the following?

- A. A page fault has occurred B. A page has corrupted data
C. A page has been modified after being loaded into cache D. An illegal access of page

isro2014 operating-system virtual-memory paging

Answer key

36.32.8 Virtual Memory: ISRO CSE 2014 | Question: 77



What is the size of the physical address space in a paging system which has a page table containing 64 entries of 11 bit each (including valid and invalid bit) and a page size of 512 bytes?

- A. 2^{11} B. 2^{15} C. 2^{19} D. 2^{20}

isro2014 virtual-memory paging

Answer key

36.32.9 Virtual Memory: ISRO CSE 2016 | Question: 22



Let the page fault service time be 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every 10^6 memory accesses, what is the effective access time for the memory?

- A. 21.4 ns B. 29.9 ns C. 23.5 ns D. 35.1 ns

operating-system virtual-memory paging isro2016

Answer key

36.32.10 Virtual Memory: ISRO CSE 2023 | Question: 57



State whether true or false:

- Virtualization facilitates multiple OS simultaneously on the same hardware
- In dual boot, multiple OS can run simultaneously

- A. (i) True (ii) False B. (i) False (ii) True
C. (i) True (ii) True D. (i) False (ii) False

isro-cse-2023 operating-system virtual-memory

Answer key

Answer Keys

36.0.1	D	36.0.2	C	36.0.3	D	36.0.4	B	36.0.5	C
36.0.6	B	36.0.7	D	36.0.8	A	36.0.9	B	36.0.10	B
36.1.1	C	36.2.1	D	36.3.1	B	36.4.1	A	36.5.1	A
36.5.2	A	36.5.3	A	36.5.4	A	36.6.1	D	36.6.2	A
36.6.3	A	36.6.4	D	36.6.5	B	36.6.6	A	36.6.7	C
36.6.8	A	36.6.9	B	36.6.10	D	36.6.11	D	36.7.1	C
36.7.2	B	36.7.3	B	36.8.1	A	36.9.1	A	36.10.1	D
36.10.2	A	36.10.3	C	36.10.4	C	36.11.1	A	36.12.1	B
36.13.1	A	36.13.2	B	36.13.3	D	36.14.1	C	36.15.1	B
36.15.2	C	36.15.3	C	36.15.4	A	36.15.5	A	36.15.6	C
36.16.1	B	36.17.1	B	36.17.2	B	36.18.1	A	36.18.2	A
36.18.3	C	36.18.4	A	36.18.5	B	36.18.6	C	36.18.7	B
36.18.8	C	36.18.9	D	36.18.10	B	36.18.11	C	36.18.12	D
36.18.13	B	36.19.1	D	36.20.1	D	36.20.2	B	36.20.3	B
36.20.4	C	36.20.5	B	36.20.6	B	36.20.7	B	36.20.8	D
36.21.1	D	36.21.2	B	36.21.3	B	36.21.4	B	36.21.5	A
36.21.6	B	36.21.7	B	36.21.8	A	36.21.9	D	36.21.10	D
36.21.11	C	36.21.12	B	36.21.13	A	36.21.14	C	36.21.15	B
36.21.16	A	36.21.17	B	36.21.18	B	36.21.19	D	36.21.20	A
36.21.21	A	36.21.22	C	36.21.23	D	36.22.1	C	36.22.2	B
36.22.3	A	36.22.4	C	36.22.5	D	36.22.6	C	36.22.7	C
36.23.1	D	36.24.1	A	36.25.1	A	36.25.2	B	36.25.3	D
36.25.4	D	36.25.5	D	36.25.6	D	36.25.7	A	36.25.8	B
36.25.9	C	36.25.10	C	36.26.1	C	36.27.1	B	36.27.2	C
36.28.1	B	36.28.2	B	36.28.3	A	36.28.4	B	36.28.5	C
36.28.6	B	36.29.1	A	36.30.1	D	36.30.2	A	36.30.3	C
36.31.1	C	36.32.1	C	36.32.2	B	36.32.3	A	36.32.4	C
36.32.5	D	36.32.6	C	36.32.7	C	36.32.8	C	36.32.9	B
36.32.10	A								